

MONALISA DOKANIA

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PROFESSIONAL SUMMARY

Founding Software Engineer with experience in full-stack development, scalable/reliable backend services, distributed systems, data-driven pipelines, and scalable cloud platform design. Curious & motivated learner, with a strong sense of ownership.

TECHNICAL SKILLS

Languages	Python, Java, Go, TypeScript, JavaScript, HTML, CSS
Frameworks / Distributed Syst	React, Node.js, tRPC, Flask, FastAPI, REST API, Spring Boot, Microservices
Cloud & DevOps	AWS, Kubernetes, Jenkins, Docker, Terraform, CI/CD, Kafka
Databases	PostgreSQL, NoSQL, Redis, MySQL, Elasticsearch, MongoDB, data modeling
Software Engineering	SDLC, Design Patterns, System Design, System Architecture, Agile
Machine Learning & AI	Feature engineering, ML model prototyping, AI-assisted workflows, LLM frameworks

TECHNICAL EXPERIENCE

Lead Software Engineer BLUSVN Technologies	June 2024 – Present Scottsdale, AZ
• Led design and end-to-end development - from design to deployment - of a React based distributed water-delivery platform as a founding engineer, owning the full product development lifecycle from requirements to production deployment. • Developed and optimized 100+ backend APIs using TypeScript (tRPC) and PostgreSQL, including identity and authentication workflows. Improved data-fetching efficiency and reduced end-to-end latency by 40%. • Implemented high performance Golang microservices leveraging concurrent worker pools and horizontal scaling, to process 5K+ orders/sec at peak load, with reliable, asynchronous execution. • Built and maintained AWS production infrastructure with Terraform and optimized Kubernetes cluster resource management by transitioning from standard Cluster Autoscaler to Karpenter, resulting in faster node provisioning & less infrastructure overhead. • Enhanced the system's security posture and implemented AWS / EKS best practices, & security-first coding practices, reducing high-severity vulnerabilities CVEs (by 20%) & increasing cluster compliance (65% to 76%.) • Built Kafka-based event-driven location workflows to support responsive UI experiences in order & delivery status tracking. • Wrote & executed unit tests, integration tests, and load tests (Python), improving system stability & reducing production issues. • Developed Python-based load testing scripts using Locust to stress-test the system under peak load, monitoring performance metrics through Prometheus and Grafana dashboards to identify bottlenecks and implement optimizations. • Collaborated closely with cross-functional developers, reviewed 200+ PRs, managed CI/CD pipelines, improved development & debugging processes through AI-assisted workflows and contributed to architecture/design discussions. • Mentored junior developers and interns, improving team productivity and engineering quality; acted as Scrum Master, and managed Agile sprints and JIRA tasks, fostering a culture of ownership and rapid problem-solving.	
Graduate Teaching Assistant (GTA/GSA) Arizona State University (SCAI)	May 2023 - May 2024 Tempe, AZ
• Supported Software Verification, Validation Testing coursework for large classes (900+ students total). • Guided students in unit testing, debugging, code reviews, software reliability and object-oriented design concepts.	

OTHER PROJECTS

Agent-Based AI System for Document & Log Analysis 🐙 <i>Python, LangChain, FastAPI, Redis RQ, Vector DB, OpenAI</i>
• Built a custom multi-agent LLM analysis pipeline (Planner → Retriever → Analyst → Critic → Reporter) to perform structured, evidence-backed document and log analysis, reducing hallucinations compared to single-prompt workflows. • Implemented a RAG workflow for semantic document retrieval and raw log diagnostics using FastAPI, Redis RQ, and Qdrant vector store, enabling asynchronous analysis of large PDFs and logs with verification and confidence scoring.

EDUCATION

Master of Science, Computer Science Arizona State University Tempe, AZ	GPA: 4.0/4.0
Bachelor of Technology, Information Technology Motilal Nehru National Institute of Technology Allahabad, UP, India	CGPA: 7.59/10.0